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Structure the Problem: Pyramids and Trees

In 2007 Amazon launched Kindle Direct Publishing (KDP) in the USA. Using this service, writers could post their e-books in Kindle format on all Amazon websites for free and price them anywhere from \$0.99 to \$200, keeping 70 percent of net revenues. KDP was an immediate success. Digital self-publishing had arrived.

Aspiring writers use such platforms to turn their manuscripts into e-books or print-on-demand books and sell them on the Internet. Digital self-publishing has helped commercialize hundreds of thousands of books that otherwise wouldn't have been. Indeed, traditional publishers receive thousands of manuscripts annually and select only a handful for publication. Despite this selectivity, they are surprisingly bad at spotting best sellers, contributing to an industry-wide fear of missing the next *Harry Potter*. Dozens of publishers rejected the little sorcerer's story before Bloomsbury finally published it.

Librinova is a start-up that successfully launched a digital self-publishing platform in France in 2014.¹ Any writer can pay to join the Librinova platform and access a range of services to edit, produce, and commercialize their book, in both print-on-demand and digital formats. Readers can buy these books through online bookstores such as Amazon. Publishers can use the platform to spot potential best sellers and target them for traditional publication. Librinova's competitive advantage is that it acts as literary agent for books that are self-published on its platform and sell over 1000 copies, helping authors to publish them conventionally. While other platforms just support writers in editing and marketing their books, Librinova sells writers the dream of being

published by an established publishing house and having their books on display in the bookshop windows of the Latin Quarter in Paris.

By 2018, over 1000 writers had joined the platform and established publishers had picked up 40 books for publication. After two successful funding rounds, investors pushed Librinova to expand internationally. Most digital platforms, such as Uber or Airbnb, pursue aggressive international expansion. The aim is to grow the platform quickly to leverage network effects, triggering a “winner takes all” dynamic in which the platform becomes a de facto standard, locking out rivals.²

Let’s suppose the founder and CEO of Librinova has asked you for advice on how to expand internationally. To tackle this issue, you need first to define and state the problem. The TOSCA framework from the previous chapter will help:

- The Trouble: shareholders are pushing for internationalization, while publishing activities are essentially domestic, with little synergies across countries. Because of linguistic and cultural differences, each nation has its own publishing industry, with local writers, readers, and publishers. International players are exceptions.
- The Owner of the problem is the founder and CEO of Librinova.
- Success can be defined as quenching the investors’ thirst for international growth in a profitable way, for example, by launching a viable self-publishing platform in a foreign country within the next 12 months.
- Constraints are strategic, financial, and organizational: self-publishing is difficult to internationalize, and the company is small, purely domestic, with limited human and financial resources.
- Other than the CEO, the key Actors are the pushy investors and potential international partners. Traditional publishers in target countries are critical: if they don’t adopt Librinova’s platform, its competitive advantage vanishes.

You might state the core question as: *Which international expansion strategy—when, where, and how—should Librinova use to respond to shareholder pressure?*

Following the 4S approach, after stating the problem, you’ll structure it in either a hypothesis-driven or an issue-driven way. Let’s explore these two approaches in sequence.

Hypothesis-Driven Problem Structuring

International expansion is not a novel problem. Many start-ups, including digital platforms that connect providers and buyers, have gone through it. The typical hurdle is that some activities tend to be domestic by nature, which requires building operations from scratch in each country. For example, in each geography, Uber must attract drivers and travelers, both of whom are locals, and perform purely local transactions. Even though the same algorithms and platform architecture can be used everywhere, cross-country synergies are weak, hampering internationalization. This issue is particularly salient in book publishing.

Librinova has examined possible internationalization strategies. One scenario is to collaborate with De Marque, a Canadian company operating in the wholesaling and distribution of digital books. De Marque and Librinova already are commercial partners: De Marque distributes Librinova e-books through online bookstores in Europe, Canada, and the USA. Librinova also knows that De Marque has considered launching its own self-publishing platform in Canada, a project that stalled due to lack of skills and technology. A joint initiative would solve that problem and leverage the partners' complementary capabilities: Librinova's capabilities in self-publishing would help De Marque diversify into a growing arena, while De Marque's connections with the local publishing industry would help Librinova create a sister platform in Canada.

This sounds like a good candidate solution for Librinova's problem. Now, your mission, if you choose to accept it, is: discover whether teaming up with De Marque in Canada is a good option. Since you see no particular reason why the CEO would be wrong, the most interesting part of the study is not to confirm the idea, but to explore it and develop some interesting insights and recommendations on how to implement it. What a relief! Instead of looking into all possible geographies (Why not go for Europe or Asia?) and all possible entry strategies (Why not look for an acquisition target in Canada or elsewhere?), you have a good starting point from which to work.

This is a typical situation. Problem owners typically pose business problems so they point to potential solutions, at least in broad terms. Most have given some thought to the problem at hand and have conducted preliminary research. Experience also plays its part: having solved similar problems in the past helps get directly to the point and identify good candidate solutions. In such contexts, the problem solver's task is to validate the solution, examine its various aspects, refine it, and recommend an action plan for implementation.

But how do you do that in practice?

Building a Hypothesis Pyramid

In this approach, you'll consider the potential solution as a hypothesis and challenge it in all its possible dimensions. Just because you have an idea does not mean it's correct—the whole point of the exercise is to validate that. To do so, you'll build a *hypothesis pyramid* from the top down. You'll start with your leading hypothesis and break it down into sub-hypotheses that are conditions for it to be true. You might also break these first-level conditions into sub-sub-hypotheses, until they are specific enough to be proven or disproven by analyses, facts, and data (Fig. 5.1). When you've performed all the required analyses, you can climb back up the pyramid and say whether the leading hypothesis is right or wrong, or more interestingly, what is right and wrong in it, and to what extent.

If you work in a team, the hypothesis pyramid is an efficient way to allocate the analytical work within the team. This chapter focuses only on how to build a hypothesis pyramid. We'll discuss how to gather the data and carry out the analyses in Chap. 7.

The Librinova case can help illustrate this approach. You can structure the problem around the hypothetical solution that the CEO is considering: Librinova should collaborate with De Marque to launch a sister platform in Canada. This candidate solution is your leading hypothesis at the top of the pyramid (Fig. 5.2). What are the conditions for it to be true?

This hypothesis combines three choices: the choice of country (Canada), the choice of entry mode (collaboration with a local partner to create a sister platform, rather than organic entry or acquisition), the choice of partner (De

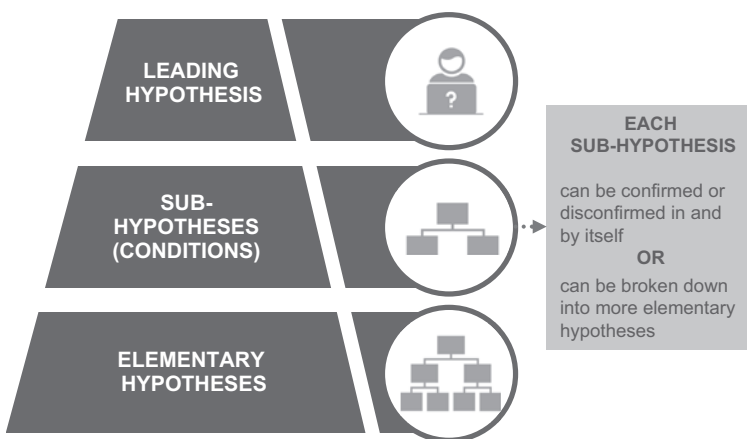


Fig. 5.1 The hypothesis pyramid



Fig. 5.2 First level of a hypothesis pyramid on the Librinova case

Marque rather than someone else). You must check if these three choices are correct: Canada must be an attractive market, collaboration on a sister platform must be the best entry mode, and De Marque must be a good partner. Otherwise, Librinova should reconsider the idea and look into another country, another entry mode, or another partner. These three sub-hypotheses are conditions that must hold (be true) for the main hypothesis to hold.

Necessary and Sufficient Conditions

The Librinova example points to a key principle in building a hypothesis pyramid: at any level of the pyramid, a hypothesis must be supported by sub-hypotheses sitting at the lower level. What we mean by “supporting” is that each sub-hypothesis is a condition for the focal hypothesis to be true. To avoid mistakes here, we must explore a distinction you may remember from your math classes between *necessary* and *sufficient* conditions.

Necessary conditions are conditions that cannot be wrong if the hypothesis is true. For example, “Socrates is mortal” is a necessary condition for “Socrates is human.” Similarly, “earning a net operating profit on sales” is a necessary condition for “achieving economic profit.” However, necessary conditions can be true while the hypothesis is wrong: “Socrates is mortal” can be true even if Socrates is not human—for example, if Socrates is your dog. Similarly, a posi-

tive operating profit is necessary to achieve economic profit, but it doesn't guarantee that the return your firm generates exceeds its cost of capital, which is the definition of economic profit. In the Librinova example, Canada must be an attractive market to justify entering it. This is a necessary condition. But Librinova could decide not to go to Canada—for example, if they realize they can't find a good partner there.

Sufficient conditions work the other way around. A sufficient condition is enough to prove a hypothesis is true, but a sufficient condition can be wrong even though the hypothesis is true: "Socrates is human" is a sufficient condition for "Socrates is mortal," but it's not a necessary condition for Socrates's mortality. Expanding a company's client base abroad is sufficient to increase its export sales, but export sales can increase for other reasons, such as variations in currency exchange rates.

You can use both types of conditions in a hypothesis pyramid, provided you know the distinction between necessary and sufficient conditions: *proving one sufficient condition right is enough to validate a hypothesis, while proving one necessary condition wrong is enough to reject it.*

From a purely logical standpoint, you'd love to find a single "killer" sufficient condition to confirm your hypothesis or a single "killer" necessary condition to reject it. In the Librinova case, the three conditions we listed are all necessary: proving one of them wrong is enough to reject the leading hypothesis. Conversely, finding a single piece of evidence that would be sufficient to demonstrate the validity of the alliance with De Marque would save you a lot of work.

In practice, however, you'll encounter two challenges. The first is that *accepting or rejecting the leading hypothesis isn't your only objective*. It's more often a means to solve a business problem as thoroughly as you can. Even if you could validate a sufficient condition that confirms the candidate solution immediately, you wouldn't stop there. You would try to find other sufficient conditions that could lead to alternative but compatible ways to solve the problem. For example, increasing sales can be achieved through growing the client base in the domestic market, increasing the average revenue per client, or penetrating new foreign markets. Each of the three conditions is sufficient to increase sales, but you would have to consider all of them and find others, if possible. The hypothesis-driven approach, if properly conducted, won't necessarily lead to a quick confirmation (or disconfirmation) of your hypothesis, but can enlarge the scope of solution options.

The second challenge is that *sufficient conditions are rare in business problems*. In the Librinova hypothesis pyramid, we don't have any sufficient conditions, but we do have a list of necessary conditions. This is typical because finding a

single sufficient condition to validate a complex plan is usually impossible. As you disaggregate the leading hypothesis into more elementary sub-hypotheses that are “searchable”—concrete and specific enough to be (in)validated by evidence—you’ll identify many conditions that are necessary, but none that is sufficient. Such necessary conditions are what detectives call “clues” and doctors “symptoms.” A red rash, especially behind the ears and on the neck, is a symptom of measles. The symptom is a necessary indicator that appears as a consequence of the disease, but it alone cannot confirm the diagnosis. Many things other than measles, such as a dermatologic disorder, can cause a red rash. More conditions are needed to ensure measles is the correct diagnosis.

For example, if you consider taking over a company, you must check at least two necessary conditions: first, that the value created by the expected synergies is significant and offsets the likely acquisition premium, and second, that the target company’s shareholders will sell their stock at such a premium. This helps you identify two sources of evidence you’ll need to obtain: (1) you’ll need to assess the potential synergies between the two firms and estimate if they’ll offset the anticipated premium, and (2) you’ll need to estimate the minimum acquisition premium which is acceptable to the target firm’s shareholders. Failing one of these two necessary conditions is enough to make the deal unattractive. They are necessary, but not sufficient.

When you list a set of necessary conditions in a hypothesis pyramid, you can’t be completely sure that validating these conditions is enough to validate the leading hypothesis. You have a nice bunch of clues, but there might be something missing. Working with a hypothesis pyramid is both a fact-finding and logical challenge: you must make sure the necessary conditions are verified and that, taken together, they are sufficient to validate the hypothesis.

The Rule of MECE

Let’s get back to the Librinova example. You must confirm all three necessary conditions to validate the leading hypothesis. But this isn’t enough. You must also check that the combination of these three necessary conditions is sufficient to validate the hypothesis. Otherwise, you might be missing another condition that could throw a wrench into the logic. As a logical requirement, the conditions should be *collectively exhaustive*.

“Collectively exhaustive” means we’ve identified all possible conditions to provide logical support for the hypothesis. A trick to make sure that a list of items is collectively exhaustive is to determine whether an “others” category

exists. If you think of categorizing car bodies into five categories (sedans, station wagons, convertibles, coupes, minivans), ask yourself whether a sixth category called “others” would be empty or not. For the categories to be collectively exhaustive, we must be able to assign any and all cars to one of the categories. Another way to check for collective exhaustiveness is to assume that all the conditions you’ve listed hold, and still try to argue against the leading hypothesis: what objections can you find? Suppose that the three conditions you’ve listed for the Librinova–De Marque alliance are confirmed. Is there another reason Librinova shouldn’t proceed? Maybe it has better options: expanding abroad might be easier in neighboring countries, such as Belgium, Germany, or Spain, rather than overseas.

In addition to being collectively exhaustive, the conditions must not overlap. In other words, they should be *mutually exclusive*. If the car body categories are mutually exclusive, then a car cannot be assigned to more than one category. The three conditions we have listed in the Librinova pyramid are mutually exclusive. Indeed, each one is self-contained and can be investigated as such. Proving any one wrong would be enough to reject the hypothesis altogether; there would be no need to look into the other conditions. Conversely, proving one right provides only a piece of support for the hypothesis, but this is a solid and independent piece of support. Once you’ve checked one condition, you can move to the next and assess whether it also holds, without conducting the same analyses twice.

The combination of these two checks—collective exhaustiveness (sufficient together) and mutual exclusivity (non-overlapping)—is often abbreviated as MECE (pronounced “missy”): mutually exclusive and collectively exhaustive. MECE is a fundamental notion and a pillar of good problem solving and solution selling. We’ll continue to refer to it throughout the book. A MECE list is like a solved jigsaw puzzle: the pieces fit with no overlap, and their combination covers the whole picture.

Checking for “MECEness” reveals that our three conditions for the Librinova–De Marque alliance are not collectively exhaustive: we didn’t compare the candidate solution with alternatives that might prove more attractive. We must add a fourth sub-hypothesis: “Collaborating with De Marque to launch a sister platform in Canada is the best available option” (Fig. 5.3). Now that we have split the leading hypothesis into four collectively exhaustive sub-hypotheses, we’ve identified four, not just three, streams of research: one on assessing the Canadian market, one on looking into the pros and cons of collaborating on a sister platform, one on evaluating De Marque as a partner, and one on alternative international expansion strategies. If you

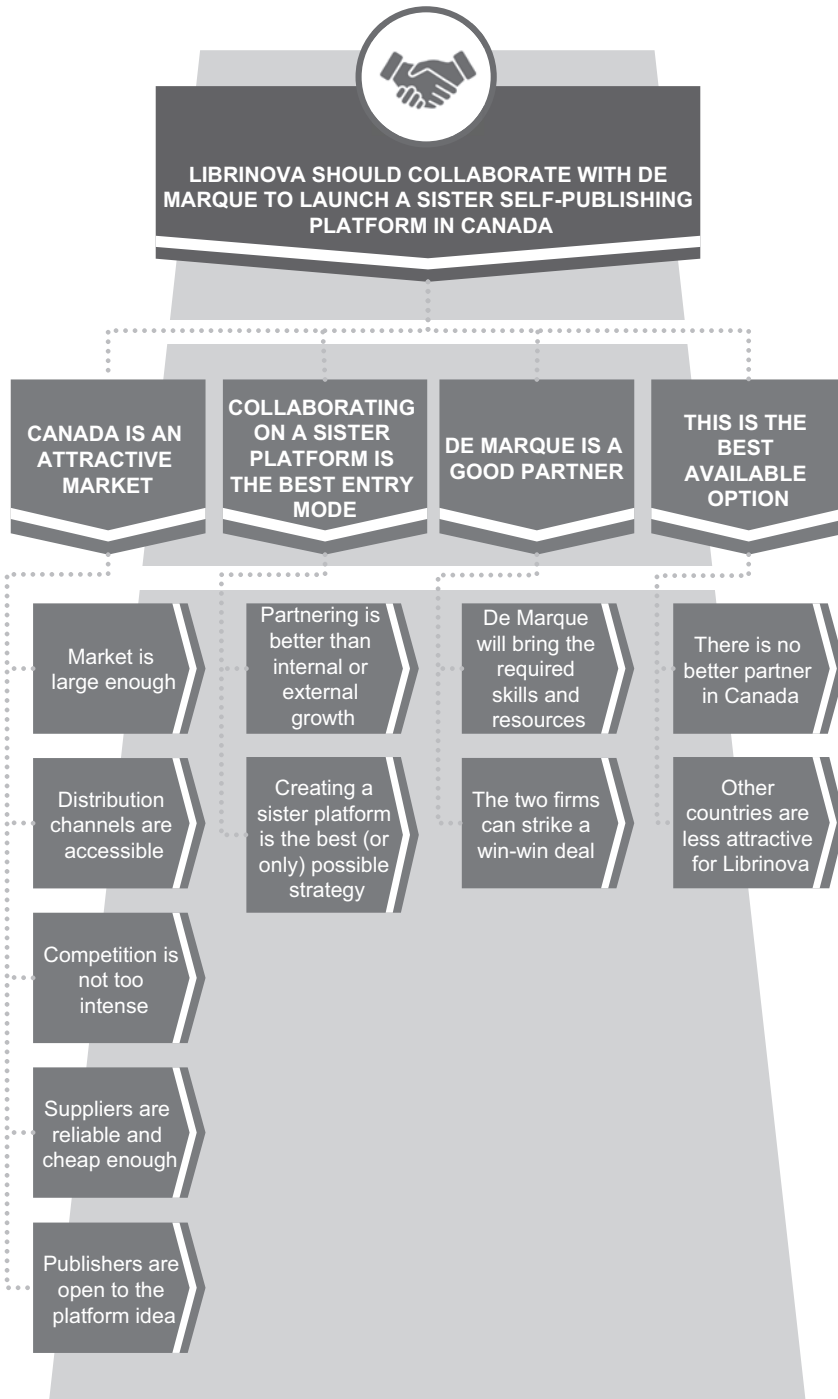


Fig. 5.3 More complete hypothesis pyramid on the Librinova case

work on a team, you can allocate these topics to different team members with little risk of overlap, because they're mutually exclusive.

The next step is to make each sub-hypothesis searchable by splitting it into elementary hypotheses you can check by collecting and analyzing data. The resulting hypothesis pyramid is depicted in Fig. 5.3.

To test the first sub-hypothesis (Canada is an attractive market), you must look into the elements that make a market attractive for Librinova. We can list five necessary conditions, based on a strategic framework called the five forces model³:

1. *The market is large enough*: assuming the same penetration rate as in France, Librinova can break even in a reasonable period of time.
2. *The distribution channels are accessible*: Librinova can put books on sale in bookstores at a reasonable cost.
3. *Competition is not too intense*: the less numerous the competitors and the higher the prices, the better.
4. *Reliable suppliers (printers, editors, web consultants, etc.) are available at a reasonable price*: this is indispensable in making the Canadian platform able to develop a range of services for writers.
5. *Canadian publishers are open to dealing with a self-publishing platform*: it is crucial for Librinova to play its literary agent role vis-à-vis local publishers, as this is its main source of competitive advantage.

These are necessary conditions: proving any of them wrong would suggest that expanding into Canada is not a good option. They are mutually exclusive, as they don't overlap and can be right or wrong independently. But are they collectively exhaustive? Since we used a framework that provides a standard approach to evaluate industry attractiveness, we probably didn't miss anything crucial: as we'll discuss in the next chapter, this is the key advantage of using a framework. However, we can't be certain we haven't overlooked something. When a decomposition is complex, it can be difficult to guarantee the conditions we identify are collectively exhaustive.

Figure 5.3 provides a decomposition of the three other "pillars" of the pyramid. In this example, the elementary hypotheses are necessary conditions, making the rule of MECE particularly important. We let the reader gauge whether our hypothesis pyramid remains MECE at all levels. We know from experience there is always room for improvement. If you work in a team, making everyone challenge the MECEness of the hypotheses, it is an excellent practice. It can prevent huge mistakes.

Logic in a hypothesis pyramid is a means to an end, rather than an end in itself. The objective is not to confirm or disconfirm the “veracity” of the leading hypothesis in purely logical terms, such as in a math proof. Instead, it’s to assess the extent to which the hypothesis holds and the relative business impact of the recommendations it suggests. In practice, you’ll engage in an iterative process through which you’ll revise and refine the leading hypothesis, based on the logic you’re developing when you structure the problem and the results you get when you perform the analyses.

If you find support for the leading hypothesis in the Librinova example, this support will probably be partial and will vary across the various “bricks” in the pyramid. This is good news: the CEO of Librinova will learn little if you just confirm her candidate solution, but she’ll learn a lot if you come up with convincing views on the expected costs and benefits of implementing it. The problem-solving process will produce interesting insights on sizing the Canadian market, assessing the competition, discovering which Canadian publishers might use the platform, shaping the agreements to propose to De Marque, and so on.

Hypothesis-Driven Problem Structuring: Pros and Cons

The Irresistible Appeal of the Hypothesis-Driven Approach

Hypothesis-driven problem solving is efficient when you start from a sound hypothesis. It saves time and energy by focusing your efforts on a candidate solution (or a range of consistent solutions). Experts and senior business people often structure problems in this way. They are frequently right, but as we’ll discuss a little later, they’re dramatically wrong sometimes.

Experts use hypotheses because their expertise provides them with a catalog of solutions to problems in their area of expertise. When experts face a problem, they start with a candidate solution that their expertise suggests—a hypothesis—and they try to validate it while adapting it to the peculiarities of the problem at hand. They recognize patterns and work from them.

Pattern recognition is a highly efficient way to solve problems. Once a pattern is recognized, the problem solver can focus on refining and quantifying the solution, to support it with a business case that makes it actionable. Like patients who want to be told how to cure their disorder, the average decision-maker is only interested in hearing about solutions and their likely impact on the problem.

The hypothesis-driven approach not only saves exploration efforts, but also anticipates solution selling. As we start with a candidate solution, the arguments we use to confirm it are also selling points we'll use to convince the problem owner of its relevance. We simultaneously generate the solution and the arguments to sell it. In the Librinova example, you can use the hypothesis pyramid to sell the solution to the CEO, who may use the same arguments to convince the investors. We'll explore this further in Chap. 10.

Finally, political and organizational circumstances also favor the hypothesis-driven approach. When a problem solver is under pressure, and admitting ignorance isn't an option, quickly identifying a candidate solution using this approach provides a shortcut between problem solving and solution selling. A CEO issuing a profit warning can't afford to say publicly he doesn't know how to address the problem. In such circumstances, you can't start from a blank sheet of paper. You must demonstrate you can think on your feet by identifying a candidate solution quickly.

For all these reasons, the problem solver's mandate is sometimes limited to confirming a pre-sold, good-enough solution that satisfies the problem owner, and to looking carefully into the conditions that will make the candidate solution work. In such contexts, it can be counterproductive to search for other options and prove you can find better ideas. For example, if Librinova's shareholders were convinced the company should go international quickly and the CEO was convinced the alliance with De Marque was the best option, would you waste your time fishing for other ideas? Investigating a pre-sold solution saves you both unsolicited investigations and solution-selling efforts. This doesn't mean you should dishonestly validate the leading hypothesis if you find it wrong, only that you can limit the scope of your work to challenging and refining it.

The Dangers of Hypothesis-Driven Problem Structuring

Hypothesis-driven problem structuring isn't without risks. It exposes you to at least five potential mistakes—the pitfalls we introduced in Chap. 2. First, hypothesis-driven thinking can creep in during the problem statement stage. You might be tempted to think like an expert in domains in which you don't have the expertise, or even in which no true expertise can be developed.⁴ You may then mistakenly recognize patterns in problems, believing, incorrectly, you know what's going on, and use these patterns to quickly arrive at candidate solutions. As we learned from the story of the music industry in Chap. 2, while this is highly efficient, it can lead one to develop hypotheses about can-

didate solutions that are flawed. At best, this results in a waste of time. At worst, it can lead to severe errors, as we illustrated with the *flawed problem definition pitfall*.

Second, once the hypothesis is stated, it immediately narrows the problem frame. Hypothesis-driven problem structuring can lead you to fall prey to a self-inflicted *narrow framing pitfall*. You focus on the solution you have in mind and look for alternatives only if your candidate solution is proved inadequate. The WYSIATI syndrome we mentioned in the introductory chapter creates the delusion that the hypothesis addresses the whole problem: nothing needs to exist outside of it.

In the Librinova example, hypothesizing that Librinova teams up with De Marque to launch a platform in Canada hinders the examination of other moves that could also solve the stated problem—how to respond to investor pressure to expand internationally. While you can consider alternative moves, as we did in Fig. 5.3, many hypothesis-driven problem solvers might limit their attention to the first three sub-hypotheses that directly relate to the candidate solution, and overlook the fourth, limiting their understanding of the problem and their search for solutions.

Moreover, even though we include a fourth sub-hypothesis, we miss out on another question: whether international expansion is an attractive strategy, given the domestic nature of the publishing industry. There might be better ways to meet shareholders' growth aspirations—for instance, by acquiring local (French) competitors or expanding into related businesses. Considering these alternatives helps redefine the problem by broadening it beyond international expansion. This is precisely the point: starting with a hypothesis encourages a narrow definition of the problem. As a consequence, if your leading hypothesis is validated, you'll never know whether a better option was feasible: "What You See Is All There Is!"

Third, the tools you use to understand and analyze the problem may implicitly limit the hypotheses you develop. This can happen when the problem owner calls on experts. The choice of experts determines the hypothesis and implicitly rules out other relevant viewpoints. For example, the CEO of an automotive company accused of tampering with emissions tests can rely on different experts to address the problem. Should he call an automotive engineer? A lawyer? A management consultant? A communications guru? All four? Each expert will have a different approach, including different frameworks endemic to their respective area of expertise, to understand the problem and formulate hypotheses. The candidate solution proposed will depend on the expert chosen and the frameworks deployed. When you pick the wrong expert, you risk facing the *wrong framework pitfall*.

Fourth, hypothesis-driven thinking can lead you to communicate the solution before solving the problem. Part of the appeal of a hypothesis pyramid is that its logic is the same as the structure of a story that sells the candidate solution. But this confusion creates risks. Would you want to see the same thinking in the White House's situation room as in its press room? In the situation room, the focus is (or should be) on finding the best solution to deal with a crisis. In the press room, the press secretary sells the solution to the press corps and the world. Identifying candidate solutions based only on what can be sold to an audience is dangerous. This is a form of the *miscommunication pitfall* we discussed in Chap. 2.

Finally, hypothesis-driven problem structuring can lead to the *solution confirmation pitfall*. A hypothesis that looks sensible can lead problem solvers to be more inclined to search for and be receptive to information that confirms it rather than information that disconfirms their belief in its soundness. Even the most experienced problem solvers can fall into this trap—the more experienced and successful they are, the greater the repertoire of cases in their mental libraries, the more likely they are to trust their instincts, increasing the risk of confirmation bias.

Logical Challenges

Another hurdle with hypothesis-driven problem structuring is that it requires more rigorous logical reasoning than meets the eye. While the approach seems user-friendly and intuitive, it creates a complex web of hypotheses and supporting conditions that risk logical mistakes.

A typical error is to confuse sufficient and necessary conditions with causes and consequences, which creates ambiguities and logical flaws. While looking for potential causes of a problem can help state and structure it, taking possible causes for hypothetical conditions can be misleading. A frequent mistake is to think that a plausible cause is a necessary condition because it's "obvious." The saying "There's no smoke without fire" illustrates this error. As fire causes smoke, we infer fire when we can see smoke. However, smoke can come from things other than fire, such as an electric boiler or an internal combustion engine. In reality, fire is sufficient, but unnecessary, to produce smoke. Therefore, the saying "There's no smoke without fire" is logically wrong. It's inherently wrong even as a metaphor because rumors are often unfounded! In a business context, many announcements, such as profit warnings, acquisition projects, or layoff plans, can cause a drop in a company's

stock price. While all are very plausible causes, none of them is a necessary condition for the stock to lose value.

In the Librinova case, a necessary condition for the Canadian market to be attractive is that it's large enough regarding the number of potential authors who would use the platform. You can check this condition by estimating the number of Canadian writers who are stuck with unpublished manuscripts. However, this necessary condition is only one reason why the Canadian market is attractive, not a cause or a consequence of its attractiveness. It is obviously not a consequence: stating that there are many unpublished writers in Canada *because* the market is attractive would be simply stupid. But, more interestingly, it is not a cause either: the statement that the number of unpublished writers *causes* the market to be more attractive is also wrong, even though it sounds more sensible. Even if Canada was packed with unpublished writers, the Canadian market could be unattractive—for example, if intense competition among existing Canadian self-publishing platforms made it structurally unprofitable.

Many strategic mistakes stem from such logical flaws in which we confuse causality and logic. How often have you heard companies announce they're entering a market *because* it's large and growing? The size and growth of a market aren't sufficient conditions for market attractiveness—limited competition is another necessary condition, plus many others.

Another enemy of logic is the misuse of correlation. Correlations can be misleading if you confuse them with logical links. Correlations aren't logical conditions. They can be spurious and illogical: while the number of drownings in swimming pools correlates with the consumption of ice cream (because both increase with summer temperature), an ice-cream prohibition will save no lives ... at least from drowning! While you can't rely on correlations at the problem-structuring phase, you can use them (with caution) as empirical evidence later in the process, when you analyze data to validate hypotheses. We'll discuss how to deal with correlational evidence in Chap. 7.

Building a hypothesis pyramid can prove difficult. The potential confusion between necessary and sufficient conditions, logic and causality, or causality and correlation can make the pyramid's logic cumbersome and fuzzy, although its visual representation makes it look clean and rational. Under its disguise of simplicity and efficiency, the hypothesis-driven approach is riddled with logical traps. Our natural tendency is to follow our intuition and "think fast," which makes us prone to fall into such traps. We tend to overinterpret correlation and take it for causation, and take causation for logical links. We may think that a list of necessary conditions is MECE and coheres into a sufficient

condition. We may see support for our hypotheses in the faintest clues, while overlooking more significant, yet disconfirming information.

Practitioners might object to our characterization of the hypothesis-driven path as risky because it's such a widely used tool in strategy consulting. When applied effectively, it's largely free of confirmation biases, logical flaws, and other pitfalls. In practice, however, when you choose a hypothesis-driven approach, you risk falling into these traps without even realizing it. A possible remedy is to proceed down the issue-driven path instead.

Issue-Driven Problem Structuring

In contrast to the hypothesis-driven approach, the issue-driven problem-structuring path might seem mundane. Its underpinning theory is much simpler. But it's less popular in business contexts, because it requires more time, thoroughness, imagination, and critical thinking.

The Cartesian Rules: MECE Strikes Again

Issue-driven problem structuring draws heavily from the Cartesian method for conducting a systematic search of the truth. We can essentially describe it by repeating the four rules that the French philosopher and mathematician René Descartes posited in his *Discourse on the Method*⁵ at the beginning of the seventeenth century:

1. Accept nothing for true without questioning it thoroughly.
2. Divide each issue into parts until you find adequate solutions to each elementary issue.
3. Conduct analyses by starting with the simplest issues and ascending step by step to reach more complex issues, while keeping a sense of order and priority, especially when considering disconnected issues.
4. Make sure nothing is omitted.

The second rule tells us that the rational way to solve a problem is to disaggregate it into issues and sub-issues, taking nothing for granted until you reach elementary issues that are simple enough to receive reliable responses. Most problems are too complex to be solved directly. Intuitive, synthetic solutions are misleading. Descartes analyzes a problem like chemists analyze a chemical, by isolating its elementary components. His method is to split each question into more detailed questions until the simple questions are

analytically searchable. Critical thinking (the first rule) is the key ingredient for the whole process.

The last two rules are what we described as the MECE principle. Elementary issues must be mutually exclusive so you can solve them one by one. You ensure nothing is omitted if all elementary points cohere into a combination that covers all the facets of the problem.

Issue-driven problem structuring consists of decomposing the problem into MECE issues and sub-issues that you can solve one by one. The practical tool to do so is the issue tree, as depicted in Fig. 5.4.

Issue Trees Versus Hypothesis Pyramids

The choice to represent hypotheses as a pyramid and issues as a tree is arbitrary. We distinguish the visual representations of the two approaches to emphasize their conceptual differences. While hypothesis pyramids consist of hypothetical statements, issue trees consist of questions. Each branch of a tree

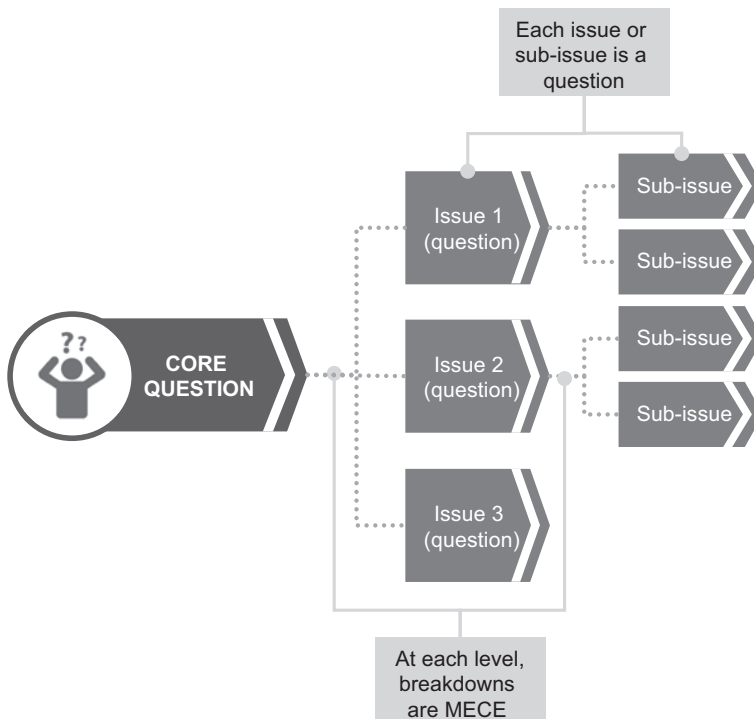


Fig. 5.4 Issue tree structure

bears a question divided into finer-grained questions on thinner branches. Thinner branches detail the broader questions posed on thicker branches to make them analytically searchable.

Although Descartes came up with his principles to guide the reasoning of individuals, consultants use issue trees as collaborative tools to organize problem-solving initiatives. They build the issue tree as a team, which fosters both critical thinking and a shared understanding of the problem structure.

In real-life situations, especially if you're under time pressure, building an issue tree may seem a waste of time. Starting with one question and ending up with more questions might seem counterproductive. You might think an issue tree is overkill because you'll never show it to the problem owner (who wants to hear about solutions, not problems). You might also believe that issue-driven structuring requires only common sense and critical thinking, as opposed to the hypothesis-driven path, which requires insight and business experience.

This is like saying issue trees are for beginners and hypothesis pyramids for advanced problem solvers, which is incorrect. Building a relevant issue tree requires insight and business sense. Experience helps a lot. An advantage of the issue-driven approach is that it's more thorough and avoids the traps induced by the hypothesis-driven approach.

Growing Issue Trees

Let's return to the Librinova example. We stated the problem as: *Which international expansion strategy—when, where, and how—should Librinova use to respond to shareholder pressure?*

To structure the problem using an issue tree, we must split this core question into sub-issues. Since we've stated the problem as a choice among possible options, one possible way to structure it is to think of a table in which the possible options appear in rows and the assessment criteria appear in columns. You could put a grade in each cell of the table to rank the different options based on the criteria. Thus, we can decompose the core question into two sub-issues:

- What are Librinova's international expansion options?
- What are the assessment criteria for Librinova and its shareholders?

Given the context, we must pay special attention to the De Marque option. A practical way to do so is to structure the first sub-issue around it. Figure 5.5 shows

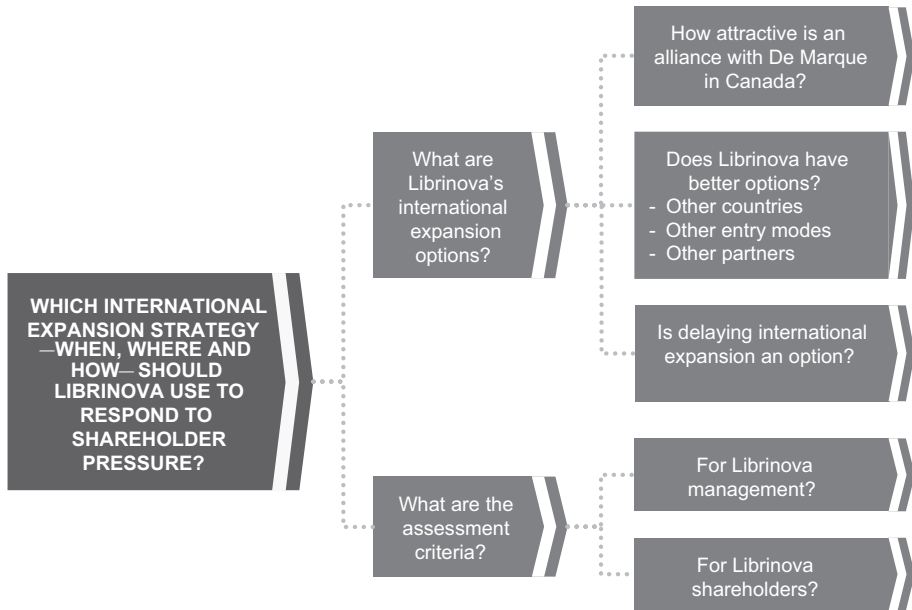


Fig. 5.5 Preliminary issue tree on the Librinova case

a preliminary issue tree. In it, we split the first sub-issue into two branches: (1) analyzing the De Marque option and (2) looking into alternatives, including other entry modes (in Canada or elsewhere), other countries, and other potential partners (in Canada or elsewhere).

For the breakdown to be MECE, we must consider the option to resist the shareholders' pressure and not to expand abroad for the moment. Librinova might have legitimate reasons to choose this route and, no matter our final recommendation, we had better examine these reasons. However, this line of investigation differs significantly from comparing international expansion options based on countries, entry modes, and potential partners. It entails analyzing Librinova's business model and the ability to replicate it in foreign countries. We think it makes sense to consider this stream of investigation as a sub-issue.

The De Marque option also deserves specific attention, so we can make it a sub-issue as well. Examining the De Marque option and discovering whether other options are available are different lines of work. For example, identifying other possible geographies draws on aggregated data on self-publishing in different countries, while analyzing the Canadian option requires more fine-grained information on local market circumstances. This leads us to the complete issue tree depicted in Fig. 5.6.⁶

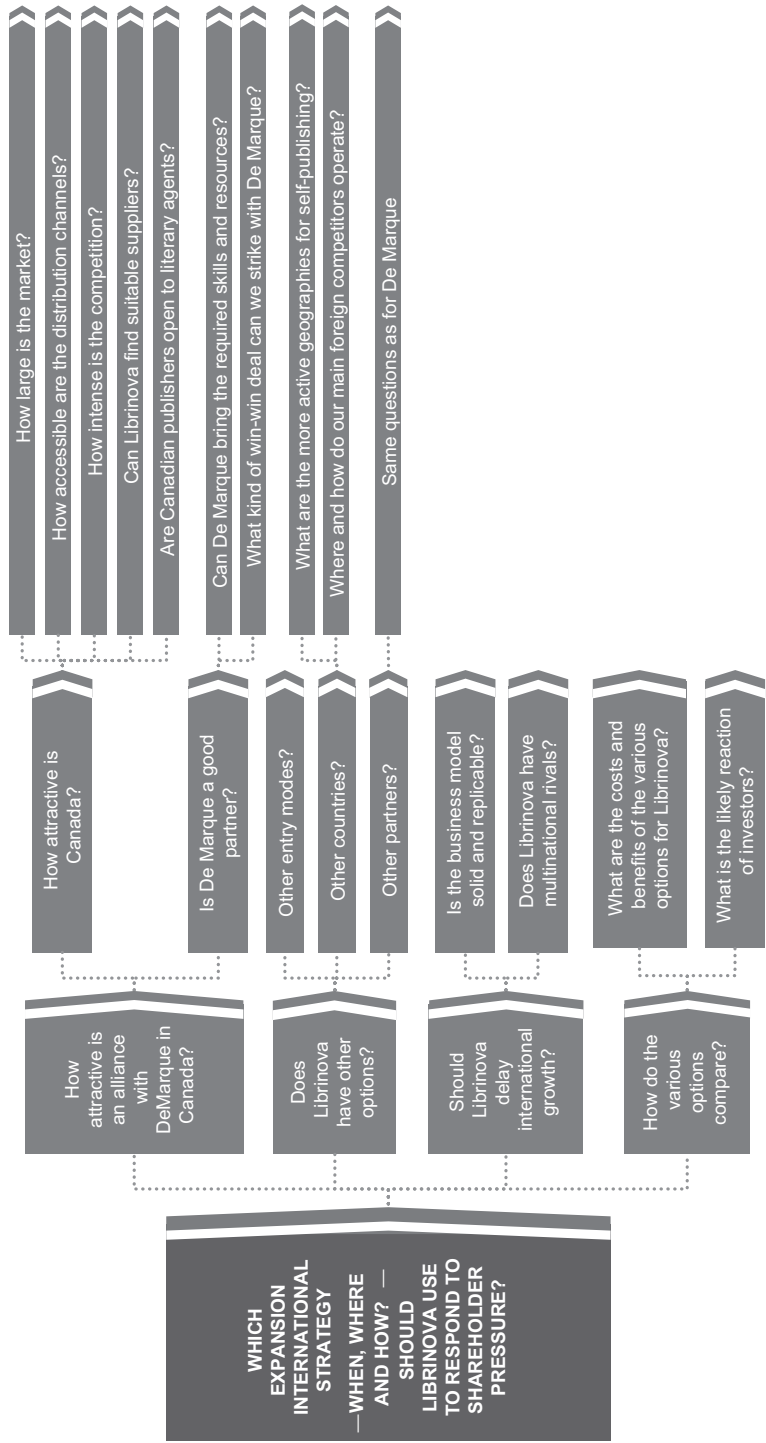


Fig. 5.6 More complete issue tree on the Librinova case

The 80/20 Rule

The challenge is to be thorough while keeping the issue tree manageable. Trying to make the issue tree exhaustive is beneficial at the beginning of the problem-solving process, but at some point, you must prune it by setting priorities and eliminating “dead-end” branches. A general rule of thumb is that 20 percent of the possible issues have the potential to generate 80 percent of the insights.

More than the percentage, which varies a lot, the message to remember is that, in most problems, a minority of key factors plays a major role. For example, in a company’s business, 20 percent of key customers often generate 80 percent of sales, or 20 percent of product lines yield 80 percent of profits. Focusing on issues that entail maximum impact is more efficient. You can drop low-impact branches at early stages and keep others in standby mode while you are examining the most impactful issues. However, the tree must be trimmed based on likely impact, not because some areas are easier than others to search!

In the Librinova example, looking into the reasons Librinova could resist shareholder pressure and not go international might be a priority. It might reveal interesting insights about the company’s strategy and shed a different light on the other sub-issues. When taking a hypothesis-driven approach, we were inclined to neglect this option (Figs. 5.2 and 5.3). Starting with an open issue and breaking it down in a MECE way can help you broaden the initial question by unearthing sub-issues you would have otherwise overlooked.

Growing a Tree or Building a Pyramid?

For narrowly defined problems, especially “yes or no” issues, the hypothesis-driven and the issue-driven approaches are equivalent. Consider, for example, a problem stated as whether or not to acquire a particular target company. An issue tree would ask the “yes or no” question, while a hypothesis pyramid would start by hypothesizing that the acquisition should be done. Both approaches would be essentially identical. The issue tree would look very much like the hypothesis pyramid, except all items would come as questions instead of hypothetical statements. The only benefit of the issue-driven approach would be to mitigate the risk of the solution confirmation pitfall, as the problem solver would be more likely to keep an open mind.

However, in most situations, you must choose between a hypothesis pyramid and an issue tree. Our recommendation here is simple: *the “default choice” should be to use an issue tree*, unless there are compelling reasons to be content with a hypothesis pyramid.

As we’ve seen, hypothesis-driven thinking is quicker and feels more natural. If you’ve never thought about how you solve problems, you probably follow the hypothesis-driven path. But the issue-driven structure can help you avoid common traps of hypothesis-driven thinking and reach more insightful conclusions, even when you have candidate solutions in mind. Whenever you have sufficient time and latitude to investigate the problem in depth, issue trees should be your first call.

Limit the use of hypothesis-driven thinking to two cases only:

1. *You have strong reasons to believe in your hypothesis.* This can be the case if you have deep expertise, or if the problem is trivially simple. In such contexts, an issue tree would be overkill. But reverse the burden of proof: before accepting any candidate solution as your leading hypothesis, put the onus on those who suggest this route and invite them to justify their belief in the hypothesis.
2. *You don’t have the luxury of building an issue tree.* This may be the case because of time constraints, or because you’re under pressure to show you have a solution in mind from the get-go (and you’re also ready to accept the existing problem definition without challenging it). It’s a risky route, but it is sometimes the only one.

Figure 5.7 summarizes the pros and cons of hypothesis pyramids and issue trees and the contexts in which you can use one or the other.

* * *

Overall, we recommend issue trees, rather than hypothesis pyramids, to structure problems. This helps mitigate confirmation biases and other problem-solving pitfalls. However, the main hurdle with issue trees is that decomposing issues and sub-issues into MECE components can be long and difficult, especially on unfamiliar problems, which are precisely the problems for which the issue-driven path is the most adequate. In the next chapter, we’ll show you how to use frameworks to overcome this obstacle.

A limitation of the two problem-structuring approaches we’ve discussed is that they’re analytical and rarely foster creativity. They tell us which analyses to conduct to find or confirm solutions based on logical reasoning, but they

	HYPOTHESIS PYRAMID	ISSUE TREE
PROS 	• Fast • Intuitive • Easy to sell	• Rigorous • Often generates new insights • Can lead to conclusive findings
CONS 	• Often too narrow (problem definition and narrow framing pitfalls) • Often misleading (solution confirmation and wrong framework pitfalls) • Almost never entirely conclusive (logical challenges)	• Time consuming • Difficult • Not a good communication tool • May not generate a solution
USE 	• When you have very strong reasons to believe in your hypothesis, because ◦ You have real expertise in the field ◦ The problem is simple enough to be “recognized” as part of a familiar pattern Or • When you cannot build an issue tree because ◦ The problem is intentionally defined in a narrow way and you do not want to challenge it ◦ You are under time pressure or need to demonstrate you have an answer	• By default, whenever you can Or • When the conditions to build a hypothesis pyramid are not met

Fig. 5.7 When to use a hypothesis pyramid or an issue tree

don't tell us how to come up with innovative solutions. If the problem requires thinking out of the box and inventing a new solution, they're both likely to fail. If you're looking to develop innovative solutions, you should turn to the design thinking approach we'll discuss in Chaps. 8 and 9.

* * *

Chapter 5 in One Page

- *Hypothesis-driven* problem structuring starts with a candidate solution to your core question, which you try to confirm or disconfirm:
 - *Librinova question: what international expansion strategy to implement?*
 - *CEO's candidate solution: team up with De Marque*
- To build a *hypothesis pyramid*, start with the candidate solution as the leading hypothesis at the top. Break it down into *sub-hypotheses*—conditions for the hypothesis to hold. If needed, break down sub-hypotheses again.
 - *At least three necessary conditions for Librinova's hypothesis: 1. Canada is an attractive market. 2. Collaboration on a sister platform is the best entry mode. 3. De Marque is a good partner.*
- At each level of the pyramid, conditions that support a hypothesis must be MECE: mutually exclusive (no overlap) and collectively exhaustive (collectively sufficient to confirm the hypothesis).
- Challenge each hypothesis thoroughly to confirm, disprove, or refine it.
- The hypothesis-driven path is efficient, leverages expertise and pattern recognition, and facilitates solution selling. But it has serious limitations:
 - Favors all five pitfalls discussed in Chap. 2
 - Creates logical challenges: necessary and sufficient conditions $\neq$ causes and consequences; correlations $\neq$ causal relations
- To take the *issue-driven path*, start with your *core question* and disaggregate it:
 - into questions, *not* statements or hypotheses;
 - in a MECE way: at each level, sub-issues must be both mutually exclusive and collectively exhaustive;
 - using the 80/20 rule (20 percent of issues = 80 percent of insights.)
 - *Librinova's core question can be disaggregated into four sub-questions: 1. How attractive is an alliance with De Marque in Canada? 2. Does Librinova have better options? 3. Should Librinova delay international expansion? 4. How do the options compare?*
- *By default, use issue trees.* Choose hypothesis pyramids only when you have strong trust in your hypothesis, or when you cannot build an issue tree.

Notes

1. Garrette, B. (2017). *Librinova: A Start-up for Digital Innovation in Publishing*. HEC Paris case study. Retrieved from <https://www.thecasecentre.org/educators/search/results?s=CEAB419890C5284D1B09BE222FD036A3>.
2. Eisenmann, T.R., Parker, G.G., & Van Alstyne, M.W. (2006). Strategies for Two-sided Markets. *Harvard Business Review*, 84(10), 92–101.
3. Porter, M.E. (1980). *Competitive Strategy*. New York: Free Press.
4. Kahneman, D., & Klein, G. (2010). Strategic Decisions: When Can You Trust Your Gut? *McKinsey Quarterly*. Retrieved from <https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/strategic-decisions-when-can-you-trust-your-gut>.
5. Descartes, R. (1637). *Discours de la Méthode*. Paris: Libro, 2013 reprint. English edition: *A Discourse on the Method*. Oxford: Oxford University Press, 2008.
6. As with all the examples provided in this book, this one is simply an illustration of how to handle an issue-driven structuring effort, not an “ideal” or “perfect” version of an issue tree. Our version might offer some room for improvement. For example, the split of the third sub-issue on delaying international expansion does not look very MECE to us. Do you have better ideas?